

A CFD-type Model of Semiconductor Dynamics for Operating PINNs in Telecom Laser EDA

Revised Version 2.2: Consistent Formula Set and Band-energy Closure

June 2026

Abstract

This document provides a cleaned and internally consistent version of the CFD-type semiconductor dynamics model. The model is intended for Physics-Informed Neural Networks (PINNs) used in Electronic Design Automation (EDA) of telecom lasers. The purpose is not to reproduce every microscopic quantum process. The purpose is to define a compact physics-informed structure that links engineering inputs, carrier transport, band-energy dynamics, photon density, electromagnetic frequency, and heat generation to telecom-relevant outputs such as optical power, output frequency, wavelength stability, temperature stability, and electrical-to-optical efficiency.

1 Motivation and modelling principle

The proposed model introduces the current velocity of semiconductor carriers as a field variable. This current velocity is different from both the traditional electrical current density and the quantum band velocity of electrons. Introducing this velocity converts the semiconductor dynamics from a classical drift-diffusion formulation into a CFD-type transport model containing advection, diffusion, source, and sink terms.

The model is motivated by telecom laser EDA. A fully mechanistic semiconductor laser model can be too complex and too expensive to verify experimentally. A pure neural-network model is easy to operate but is strongly restricted by data availability and may carry high uncertainty. A PINN-based model is used as a compromise: the governing equations provide physics-informed structure, while experimental data calibrate uncertain closure parameters.

From the EDA perspective, the most important outputs are not all microscopic states. The important outputs are stable and predictable performance metrics under given design and operating inputs:

$$P_{\text{opt}}, \quad f_{\text{out}}, \quad \lambda_{\text{out}}, \quad \Delta f, \quad T_{\text{max}}, \quad \eta_{\text{EO}}.$$

2 State variables

The revised model uses the following state variables.

Symbol	Meaning	Role in EDA model
$\psi(\mathbf{x}, t)$	electric potential	electrical forcing
$\mathbf{E}_0 = -\nabla\psi$	quasi-static electric field	carrier driving field
$n(\mathbf{x}, t)$	electron density	carrier population
$p(\mathbf{x}, t)$	hole density	carrier population
$\mathbf{V}_n(\mathbf{x}, t)$	electron current velocity	CFD-type transport velocity
$\mathbf{V}_p(\mathbf{x}, t)$	hole current velocity	CFD-type transport velocity
$e(\mathbf{x}, t)$	effective transition band energy	determines photon frequency
$T(\mathbf{x}, t)$	lattice or ambient temperature	thermal stability and band-energy shift
$s(\mathbf{x}, t)$	photon density	optical energy population
$\mathbf{E}(\mathbf{x})$	optical electric field phasor	electromagnetic wave output

The scalar e is the effective electron-hole transition energy. It is introduced mainly to calculate the electromagnetic frequency of emitted photons. It should be interpreted as an effective, averaged, temperature-dependent band-energy variable, not as a directly observable microscopic particle state.

3 Electrostatic constraint

For a quasi-static semiconductor device, the electric potential is governed by Poisson's equation:

$$-\nabla \cdot (\varepsilon \nabla \psi) = q (p - n + N_D^+ - N_A^-). \quad (1)$$

Equivalently, in a normalized form:

$$-\nabla \cdot \left(\frac{\varepsilon}{q} \nabla \psi \right) = p - n + N_D^+ - N_A^-. \quad (2)$$

Here ε is the permittivity, q is the elementary charge, N_D^+ is ionized donor density, and N_A^- is ionized acceptor density. Equation (1) replaces the time-dependent potential expression in the earlier draft. This is more consistent for a device-level quasi-static electrical field used to drive carrier transport.

4 Carrier mass conservation

Let G_n, G_p be generation rates and R_n, R_p be recombination rates. For a compact EDA model, they may be treated as calibrated source/sink rates.

The electron density equation is:

$$\frac{\partial n}{\partial t} + \nabla \cdot (n \mathbf{V}_n) = \nabla \cdot (D_n \nabla n) + G_n - R_n. \quad (3)$$

The hole density equation is:

$$\frac{\partial p}{\partial t} + \nabla \cdot (p \mathbf{V}_p) = \nabla \cdot (D_p \nabla p) + G_p - R_p. \quad (4)$$

If G and R are defined as first-order rates with unit s^{-1} , then equations (3) and (4) can be written as:

$$\frac{\partial n}{\partial t} + \nabla \cdot (n \mathbf{V}_n) = \nabla \cdot (D_n \nabla n) + (G_n - R_n)n, \quad (5)$$

$$\frac{\partial p}{\partial t} + \nabla \cdot (p \mathbf{V}_p) = \nabla \cdot (D_p \nabla p) + (G_p - R_p)p. \quad (6)$$

To avoid ambiguity, this document uses G_n, G_p, R_n, R_p as volumetric source terms unless otherwise stated.

Temperature-dependent diffusivities may be written as:

$$D_n(T) = D_{n0} + \alpha_n T, \quad D_p(T) = D_{p0} + \alpha_p T. \quad (7)$$

5 Current velocity dynamics

The current velocities are kept as independent CFD-type variables. They are not replaced by classical drift-diffusion velocities.

A weak incompressibility constraint can be used as a reduced-order closure:

$$\nabla \cdot \mathbf{V}_n = 0, \quad \nabla \cdot \mathbf{V}_p = 0. \quad (8)$$

The electron current velocity equation is:

$$\frac{\partial \mathbf{V}_n}{\partial t} + (\mathbf{V}_n \cdot \nabla) \mathbf{V}_n = -\kappa_n \nabla \psi + \kappa_{en} \nabla e + \nabla \cdot (D_{vn} \nabla \mathbf{V}_n) - \gamma_{vn} \mathbf{V}_n + \mathbf{S}_{vn}. \quad (9)$$

The hole current velocity equation is:

$$\frac{\partial \mathbf{V}_p}{\partial t} + (\mathbf{V}_p \cdot \nabla) \mathbf{V}_p = +\kappa_p \nabla \psi - \kappa_{ep} \nabla e + \nabla \cdot (D_{vp} \nabla \mathbf{V}_p) - \gamma_{vp} \mathbf{V}_p + \mathbf{S}_{vp}. \quad (10)$$

Here κ_n, κ_p control electrical forcing, κ_{en}, κ_{ep} control band-energy-gradient forcing, D_{vn}, D_{vp} are velocity diffusivities, and γ_{vn}, γ_{vp} are damping coefficients. The source terms $\mathbf{S}_{vn}, \mathbf{S}_{vp}$ can be calibrated by PINNs if needed.

6 Temperature-dependent band-energy dynamics

6.1 Physical interpretation

The effective band energy $e(\mathbf{x}, t)$ is used to predict the emitted electromagnetic frequency. For a specific material, the primary engineering dependency of the band energy is temperature. A higher temperature usually lowers the effective band gap, which shifts the output frequency downward and wavelength upward.

The local thermal-equilibrium band energy is denoted by:

$$e_{\text{eq}}(T). \quad (11)$$

A simple linear closure is:

$$e_{\text{eq}}(T) = e_{\text{ref}} - \chi_T (T - T_{\text{ref}}), \quad \chi_T > 0. \quad (12)$$

A more classical empirical closure is the Varshni-type relation:

$$e_{\text{eq}}(T) = E_g(0) - \frac{\alpha_g T^2}{T + \beta_g}. \quad (13)$$

For PINN-based EDA, equation (12) is often sufficient at the first stage. Equation (13) can be used when measured material data are available.

6.2 Revised band-energy equation

The earlier draft treated band energy as a transported conservation quantity with electric work and recombination source/sink terms. The revised version keeps transport and diffusion, but adds a clear relaxation mechanism toward the local temperature-dependent equilibrium value.

The recommended equation is:

$$\frac{\partial e}{\partial t} + \mathbf{V}_e \cdot \nabla e = \nabla \cdot (D_e \nabla e) + \eta_E (\mathbf{V}_p - \mathbf{V}_n) \cdot \nabla \psi - \frac{e - e_{\text{eq}}(T)}{\tau_e} + S_{e,GR}. \quad (14)$$

The effective transport velocity of band energy can be defined as:

$$\mathbf{V}_e = w_n \mathbf{V}_n + w_p \mathbf{V}_p, \quad w_n + w_p = 1. \quad (15)$$

A simple symmetric choice is:

$$w_n = w_p = \frac{1}{2}, \quad \mathbf{V}_e = \frac{\mathbf{V}_n + \mathbf{V}_p}{2}. \quad (16)$$

The relaxation term:

$$-\frac{e - e_{\text{eq}}(T)}{\tau_e} \quad (17)$$

means that the band energy does not fluctuate freely. It is restored toward the local temperature-controlled equilibrium band energy. This term is the key improvement for making the model more physically interpretable and numerically stable.

The electric-field perturbation term:

$$\eta_E(\mathbf{V}_p - \mathbf{V}_n) \cdot \nabla \psi \quad (18)$$

represents a calibrated fraction of electrical work that perturbs the effective transition energy. The remaining electrical work should be converted into heat.

The generation/recombination contribution can be written as:

$$S_{e,GR} = \eta_G G_e (e_{\text{eq}}(T) - e) - \eta_R R_e e. \quad (19)$$

Here G_e and R_e are effective generation and recombination intensities, while η_G and η_R are calibration factors. This expression is easier to interpret than a simple $(G - R)e$ term because it separates newly generated carrier energy from energy removed by recombination.

7 Photon density dynamics

Photons are described by density and energy, not by a detailed mode expansion. The photon density is denoted by $s(\mathbf{x}, t)$.

The photon density equation is:

$$\frac{\partial s}{\partial t} = \nabla \cdot (D_s \nabla s) + G_\gamma - A_\gamma - L_\gamma. \quad (20)$$

A compact closure is:

$$\frac{\partial s}{\partial t} = \nabla \cdot (D_s \nabla s) + b_\gamma R_{\text{rad}} - \alpha_{\text{abs}} s - \frac{s}{\tau_\gamma}. \quad (21)$$

Here D_s is the effective photon dispersion coefficient, b_γ converts radiative recombination into photon density, α_{abs} is the absorption coefficient, and τ_γ is the effective photon lifetime or escape time.

For a reduced-order EDA model, photon spatial propagation may be simplified to a lumped equation:

$$\frac{ds}{dt} = b_\gamma R_{\text{rad}} - \alpha_{\text{abs}} s - \frac{s}{\tau_\gamma}. \quad (22)$$

8 Photon energy, frequency, and wavelength

The emitted photon energy is determined by the effective transition band energy:

$$e_\gamma = \eta_\gamma e, \quad 0 < \eta_\gamma \leq 1. \quad (23)$$

The ordinary optical frequency is:

$$f_\gamma = \frac{e_\gamma}{h} = \frac{\eta_\gamma e}{h}. \quad (24)$$

The angular frequency used in the frequency-domain electromagnetic equation is:

$$\Omega_\gamma = 2\pi f_\gamma = \frac{e_\gamma}{\hbar}. \quad (25)$$

The wavelength in a medium with refractive index n_r is:

$$\lambda_\gamma = \frac{c}{n_r f_\gamma} = \frac{ch}{n_r \eta_\gamma e}. \quad (26)$$

This notation avoids confusion between ω as an energy-related frequency and Ω as the angular frequency in Maxwell's equations.

The photon energy density is:

$$W_\gamma = se_\gamma = s\eta_\gamma e. \quad (27)$$

9 Heat conservation

The temperature equation should account for heat generated by electrical work, non-radiative recombination, photon absorption, and photon decay.

A consistent heat equation is:

$$\frac{\partial T}{\partial t} + \mathbf{V}_T \cdot \nabla T = \nabla \cdot (D_T \nabla T) + Q_{E \rightarrow T} + Q_{R \rightarrow T} + Q_{\gamma \rightarrow T} - Q_{\text{cool}}. \quad (28)$$

A practical closure is:

$$\frac{\partial T}{\partial t} + \mathbf{V}_T \cdot \nabla T = \nabla \cdot (D_T \nabla T) + \beta_T (\mathbf{V}_n - \mathbf{V}_p) \cdot \nabla \psi + a_R R_{\text{nonrad}} + \alpha_{\text{abs}} se_\gamma + \frac{se_\gamma}{\tau_\gamma} - Q_{\text{cool}}. \quad (29)$$

Here:

- β_T is the fraction of electrical work converted into heat;
- $a_R R_{\text{nonrad}}$ is heat generated by non-radiative recombination;
- $\alpha_{\text{abs}} se_\gamma$ is heat generated by photon absorption;
- se_γ/τ_γ is heat or loss associated with photon decay, depending on the boundary interpretation;
- Q_{cool} represents heat removal through the package or heat sink.

For simple thermal modelling:

$$Q_{\text{cool}} = \frac{T - T_{\text{sink}}}{\tau_T}. \quad (30)$$

10 Frequency-domain electromagnetic wave equation

The optical electric field phasor satisfies:

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) - \Omega_\gamma^2 \varepsilon \mathbf{E} = i\Omega_\gamma \mathbf{J}_\gamma. \quad (31)$$

The equivalent source current can be parameterized by radiative recombination:

$$\mathbf{J}_\gamma = J_0 R_{\text{rad}} \mathbf{d}_{\text{eff}}. \quad (32)$$

Here $\mathbf{d}_{\text{eff}}$ is an effective emission direction or polarization vector. If isotropic emission is assumed, three orthogonal components can be used:

$$\mathbf{d}_{\text{eff}} \in \{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}. \quad (33)$$

The electromagnetic wave frequency is linked to the band-energy equation by:

$$\Omega_\gamma = \frac{\eta_\gamma e}{\hbar}. \quad (34)$$

11 Boundary conditions

For PINN training and numerical simulation, boundary conditions should be stated explicitly.

11.1 Electrical boundary conditions

At electrical contacts:

$$\psi = \psi_{\text{contact}}. \quad (35)$$

At insulating boundaries:

$$\mathbf{n}_b \cdot \varepsilon \nabla \psi = 0. \quad (36)$$

11.2 Carrier boundary conditions

At injection contacts, prescribe carrier fluxes:

$$\mathbf{n}_b \cdot (n \mathbf{V}_n - D_n \nabla n) = \Phi_n^{\text{in}}, \quad (37)$$

$$\mathbf{n}_b \cdot (p \mathbf{V}_p - D_p \nabla p) = \Phi_p^{\text{in}}. \quad (38)$$

At blocking boundaries:

$$\mathbf{n}_b \cdot (n \mathbf{V}_n - D_n \nabla n) = 0, \quad (39)$$

$$\mathbf{n}_b \cdot (p \mathbf{V}_p - D_p \nabla p) = 0. \quad (40)$$

11.3 Velocity boundary conditions

At solid blocking boundaries:

$$\mathbf{V}_n \cdot \mathbf{n}_b = 0, \quad \mathbf{V}_p \cdot \mathbf{n}_b = 0. \quad (41)$$

At injection contacts, prescribe incoming current velocities or use flux conditions:

$$\mathbf{V}_n = \mathbf{V}_{n,\text{in}}, \quad \mathbf{V}_p = \mathbf{V}_{p,\text{in}}. \quad (42)$$

11.4 Band-energy boundary conditions

At thermalized contacts:

$$e = e_{\text{eq}}(T_{\text{contact}}). \quad (43)$$

At insulated material boundaries:

$$\mathbf{n}_b \cdot \nabla e = 0. \quad (44)$$

11.5 Photon boundary conditions

At reflective boundaries:

$$s_{\text{out}} = R_\gamma s_{\text{in}}. \quad (45)$$

At open optical output boundaries:

$$\mathbf{n}_b \cdot (D_s \nabla s) = k_{\text{out}} s. \quad (46)$$

At absorbing boundaries:

$$s = 0 \quad \text{or} \quad \mathbf{n}_b \cdot (D_s \nabla s) = k_{\text{abs}} s. \quad (47)$$

11.6 Thermal boundary conditions

At a heat sink:

$$T = T_{\text{sink}}. \quad (48)$$

At a convective boundary:

$$-D_T \nabla T \cdot \mathbf{n}_b = h_c (T - T_{\text{amb}}). \quad (49)$$

At an insulated boundary:

$$\mathbf{n}_b \cdot \nabla T = 0. \quad (50)$$

12 PINN training formulation

The model is designed for experimental-data-assisted PINN operation. Let the input vector be:

$$\mathbf{z} = [I_{\text{in}}, V_{\text{in}}, T_{\text{sink}}, L_{\text{cav}}, W_{\text{cav}}, H_{\text{cav}}, R_1, R_2, \text{material parameters}]. \quad (51)$$

The target output vector is:

$$\mathbf{y} = [P_{\text{opt}}, f_{\text{out}}, \lambda_{\text{out}}, \Delta f, T_{\text{max}}, \eta_{\text{EO}}]. \quad (52)$$

A PINN loss function can be written as:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{phys}} \mathcal{L}_{\text{phys}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}} + \lambda_{\text{stable}} \mathcal{L}_{\text{stable}}. \quad (53)$$

The data loss is:

$$\mathcal{L}_{\text{data}} = \|\mathbf{y}_{\text{pred}} - \mathbf{y}_{\text{exp}}\|^2. \quad (54)$$

The physics loss is formed from the residuals of equations (1), (3), (4), (9), (10), (14), (21), (29), and (31).

The stability loss can be written as:

$$\mathcal{L}_{\text{stable}} = w_f \text{Var}(f_{\text{out}}) + w_T \text{Var}(T) + w_P \text{Var}(P_{\text{opt}}). \quad (55)$$

13 EDA operation

Once trained, the model can be used as a design automation tool:

$$\mathbf{y} = \mathcal{S}_{\text{PINN}}(\mathbf{z}). \quad (56)$$

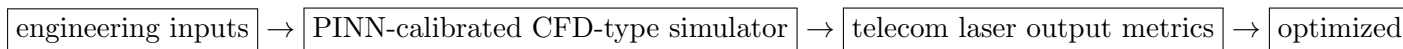
The design objective can be defined as:

$$J = w_1 (P_{\text{opt}} - P_{\text{target}})^2 + w_2 (f_{\text{out}} - f_{\text{target}})^2 + w_3 (\Delta f)^2 + w_4 (T_{\text{max}} - T_{\text{limit}})_+^2 - w_5 \eta_{\text{EO}}. \quad (57)$$

The optimized design is:

$$\mathbf{z}_{\text{design}}^* = \arg \min_{\mathbf{z}_{\text{design}}} J. \quad (58)$$

This provides the intended EDA workflow:



14 Summary of corrected formula set

The revised model is summarized as follows:

$$-\nabla \cdot (\varepsilon \nabla \psi) = q(p - n + N_D^+ - N_A^-), \quad (59)$$

$$\frac{\partial n}{\partial t} + \nabla \cdot (n \mathbf{V}_n) = \nabla \cdot (D_n \nabla n) + G_n - R_n, \quad (60)$$

$$\frac{\partial p}{\partial t} + \nabla \cdot (p \mathbf{V}_p) = \nabla \cdot (D_p \nabla p) + G_p - R_p, \quad (61)$$

$$\frac{\partial \mathbf{V}_n}{\partial t} + (\mathbf{V}_n \cdot \nabla) \mathbf{V}_n = -\kappa_n \nabla \psi + \kappa_{en} \nabla e + \nabla \cdot (D_{vn} \nabla \mathbf{V}_n) - \gamma_{vn} \mathbf{V}_n + \mathbf{S}_{vn}, \quad (62)$$

$$\frac{\partial \mathbf{V}_p}{\partial t} + (\mathbf{V}_p \cdot \nabla) \mathbf{V}_p = +\kappa_p \nabla \psi - \kappa_{ep} \nabla e + \nabla \cdot (D_{vp} \nabla \mathbf{V}_p) - \gamma_{vp} \mathbf{V}_p + \mathbf{S}_{vp}, \quad (63)$$

$$\frac{\partial e}{\partial t} + \mathbf{V}_e \cdot \nabla e = \nabla \cdot (D_e \nabla e) + \eta_E (\mathbf{V}_p - \mathbf{V}_n) \cdot \nabla \psi - \frac{e - e_{\text{eq}}(T)}{\tau_e} + S_{e,GR}, \quad (64)$$

$$\frac{\partial s}{\partial t} = \nabla \cdot (D_s \nabla s) + b_\gamma R_{\text{rad}} - \alpha_{\text{abs}} s - \frac{s}{\tau_\gamma}, \quad (65)$$

$$\frac{\partial T}{\partial t} + \mathbf{V}_T \cdot \nabla T = \nabla \cdot (D_T \nabla T) + \beta_T (\mathbf{V}_n - \mathbf{V}_p) \cdot \nabla \psi + a_R R_{\text{nonrad}} + \alpha_{\text{abs}} s e_\gamma + \frac{s e_\gamma}{\tau_\gamma} - Q_{\text{cool}}, \quad (66)$$

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) - \Omega_\gamma^2 \varepsilon \mathbf{E} = i \Omega_\gamma \mathbf{J}_\gamma, \quad (67)$$

$$f_\gamma = \frac{\eta_\gamma e}{h}, \quad \Omega_\gamma = \frac{\eta_\gamma e}{\hbar}, \quad \lambda_\gamma = \frac{ch}{n_r \eta_\gamma e}. \quad (68)$$

This version makes the role of band energy explicit: it is a transported and temperature-relaxed effective transition energy used to predict telecom laser frequency. It also separates photon density from photon frequency and links both to heat generation and electromagnetic wave output.